

Ex.no:8	MINI PROJECT – HEALTHCARE DATA ANALYTICS

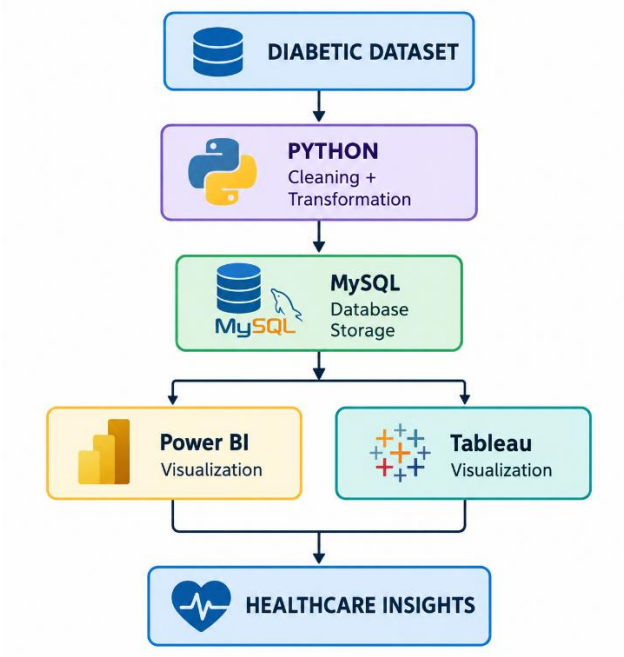
Aim:

To clean, transform, store, and analyze diabetic patient healthcare data using Python, MySQL, Power BI, and Tableau to identify patterns in patient readmissions, hospital stays, age groups, diabetes management, medications, laboratory results, and patient demographics.

Algorithm:

1. Load the diabetic patient dataset from the CSV file using Pandas.
2. Clean the data by handling missing values, removing duplicate records, standardizing column names, and correcting data types.
3. Transform the data by creating useful fields such as age groups, readmission categories, hospital stay categories, and medication status.
4. Store the cleaned and transformed dataset in a MySQL database named healthcare_analytics.
5. Connect the cleaned healthcare dataset to Power BI and Tableau for data analysis and visualization.
6. Create DAX measures and calculated fields for key metrics such as total encounters, readmitted encounters, readmission rate, average hospital stay, and average medications.
7. Develop interactive visualizations to analyze readmission patterns, age groups, gender, A1C results, hospital stay, medications, and admission types.
8. Implement KPIs, cards, slicers, filters, drill-down, drill-through, parameters, and interactive dashboard features in Power BI and Tableau.
9. Generate meaningful healthcare insights from the Power BI and Tableau dashboards.
10. End the process.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv("diabetic_data.csv")
```

```
# Create copy for processing
```

```
diabetic = df.copy()
```

```
print("Original Shape:", diabetic.shape)
```

```
diabetic.drop_duplicates(inplace=True)
```

```
# In this dataset, missing values are often represented
```

```
# using '?' rather than NaN
```

```
diabetic.replace("?", np.nan, inplace=True)
```

```
categorical_columns = [  
    "race",  
    "gender",  
    "weight",  
    "payer_code",  
    "medical_specialty",  
    "diag_1",  
    "diag_2",  
    "diag_3",  
    "max_glu_serum",  
    "A1Cresult"  
]
```

```
for col in categorical_columns:  
    if col in diabetic.columns:  
        diabetic[col] = diabetic[col].fillna("Unknown")
```

```
numeric_columns = [  
    "time_in_hospital",  
    "num_lab_procedures",  
    "num_procedures",  
    "num_medications",  
    "number_outpatient",  
    "number_emergency",  
    "number_inpatient",  
    "number_diagnoses"  
]
```

```

for col in numeric_columns:
    if col in diabetic.columns:
        diabetic[col] = pd.to_numeric(
            diabetic[col],
            errors="coerce"
        )

        diabetic[col] = diabetic[col].fillna(
            diabetic[col].median()
        )

diabetic["readmission_status"] = diabetic["readmitted"].apply(
    lambda x: "Not Readmitted"
    if x == "NO"
    else "Readmitted"
)

diabetic["readmission_category"] = diabetic["readmitted"].map({
    "NO": "Not Readmitted",
    "<30": "Within 30 Days",
    ">30": "After 30 Days"
})

age_mapping = {
    "[0-10)": "0-10",
    "[10-20)": "10-20",
    "[20-30)": "20-30",
    "[30-40)": "30-40",
    "[40-50)": "40-50",
    "[50-60)": "50-60",
    "[60-70)": "60-70",

```

```

    "[70-80)": "70-80",
    "[80-90)": "80-90",
    "[90-100)": "90+"
}

diabetic["age_group"] = diabetic["age"].map(age_mapping)

diabetic["stay_category"] = pd.cut(
    diabetic["time_in_hospital"],
    bins=[0, 3, 6, 10, 20, 100],
    labels=[
        "1-3 Days",
        "4-6 Days",
        "7-10 Days",
        "11-20 Days",
        "20+ Days"
    ]
)

diabetic["diabetes_med_status"] = diabetic[
    "diabetesMed"
].map({
    "Yes": "Medication Used",
    "No": "No Medication"
})

diabetic.columns = (
    diabetic.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

```

```
print("Cleaned Shape:", diabetic.shape)
```

```
print("\nColumns:")
```

```
print(diabetic.columns.tolist())
```

```
print("\nFirst 5 Records:")
```

```
print(diabetic.head())
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

MySQL Command Prompt

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL

```
CREATE DATABASE IF NOT EXISTS healthcare_analytics;
```

```
USE healthcare_analytics;
```

Python/Jupyter

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(
```

```
    f"mysql+pymysql://{username}:{password}@{host}:{port}/financial_analytics"
```

```
)
```

```
# Load cleaned data into MySQL
```

```
diabetic.to_sql(
```

```
    "diabetic_patients",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
    index=False,
```

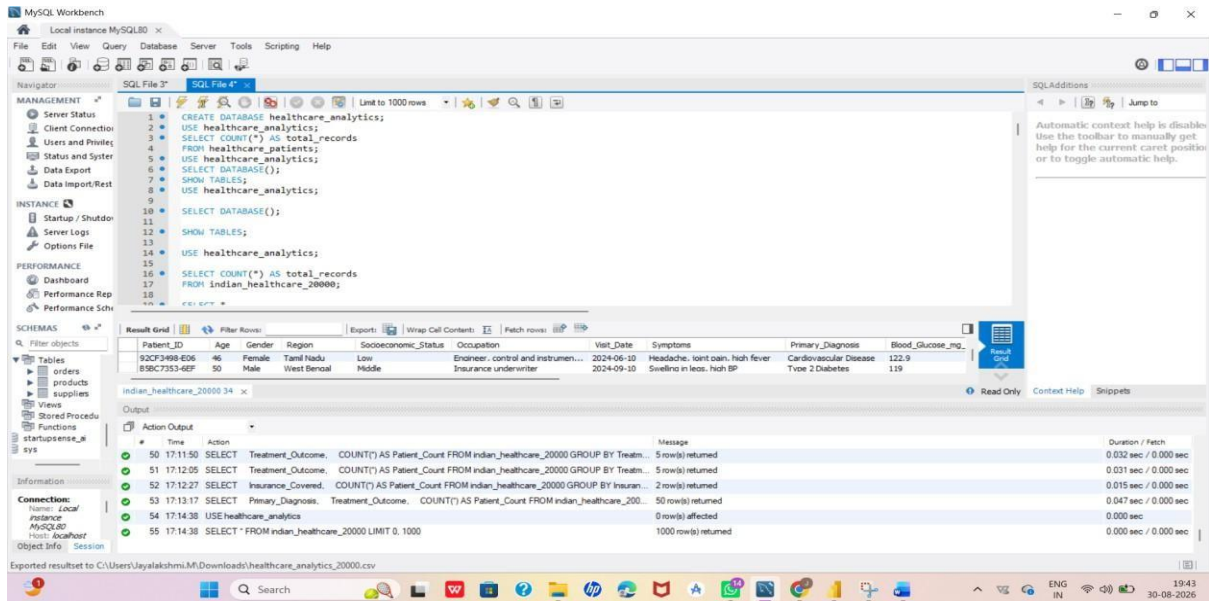
chunksize=5000

)

print("Healthcare data loaded successfully.")

Verify

pd.read_sql("SELECT COUNT(*) AS total_rows FROM diabetic_patients", db_engine)



3. Power BI – DAX Measures

Total Encounters =

COUNTROWS(diabetic_data)

Readmitted Encounters =

CALCULATE(

COUNTROWS(diabetic_data),

diabetic_data[readmitted] <> "NO"

)

Readmission Rate =

DIVIDE(

[Readmitted Encounters],

[Total Encounters],

0

)

Average Hospital Stay =

AVERAGE(diabetic_data[time_in_hospital])

Total Medications =

SUM(diabetic_data[num_medications])

Average Medications =

AVERAGE(diabetic_data[num_medications])

Average Medications =

AVERAGE(diabetic_data[num_medications])

4. Power Bi Dashboard

Created visuals in Power Bi in separate sheets and created a dashboard

Output:

Original Shape: (101766, 50)

Cleaned Shape: (101766, 55)

Columns:

['encounter_id', 'patient_nbr', 'race', 'gender', 'age', 'weight', 'admission_type_id', 'discharge_

First 5 Records:

	encounter_id	patient_nbr	race	gender	age	weight \
0	2278392	8222157	Caucasian	Female	[0-10)	Unknown
1	149190	55629189	Caucasian	Female	[10-20)	Unknown
2	64410	86047875	AfricanAmerican	Female	[20-30)	Unknown
3	500364	82442376	Caucasian	Male	[30-40)	Unknown
4	16680	42519267	Caucasian	Male	[40-50)	Unknown

```
[10]: pd.read_sql(
      "SELECT COUNT(*) AS total_rows FROM healthcare_records",
      db_engine
    )
```

```
[10]: total_rows
0    101766
```

```
[11]: pd.read_sql(
      "SELECT * FROM healthcare_records LIMIT 5",
      db_engine
    )
```

```
[11]: encounter_id  patient_nbr      race  gender  age  weight  admission_type_id  discharge_disposition_id  admission_source_id  time_in_hospital  num_lab_procedures  num_procedures  num_medications  number_outpatient
0      2278392    8222157    Caucasian  Female  [0-10)  ?         6              25              1              1              41              0              1              0
1      149190    55629189    Caucasian  Female  [10-20)  ?         1              1              7              3              59              0              18              0
2       64410    86047875  AfricanAmerican  Female  [20-30)  ?         1              1              7              2              11              5              13              2
3      500364    82442376    Caucasian  Male    [30-40)  ?         1              1              7              2              44              1              16              0
4       16680    42519267    Caucasian  Male    [40-50)  ?         1              1              7              1              51              0              8              0
```

5 rows x 47 columns



diabetic_data.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id
2278392	8222157	Caucasian	Female	[0-10]	?	6	25	
149190	55629189	Caucasian	Female	[10-20]	?	1	1	
64410	86047875	AfricanAmerican	Female	[20-30]	?	1	1	
500364	82442376	Caucasian	Male	[30-40]	?	1	1	
16680	42519267	Caucasian	Male	[40-50]	?	1	1	
35754	82637451	Caucasian	Male	[50-60]	?	2	1	
55842	84259809	Caucasian	Male	[60-70]	?	3	1	
63768	114882984	Caucasian	Male	[70-80]	?	1	1	
12522	48330783	Caucasian	Female	[80-90]	?	2	1	
15738	63555939	Caucasian	Female	[90-100]	?	3	3	
28236	89869032	AfricanAmerican	Female	[40-50]	?	1	1	
36900	77391171	AfricanAmerican	Male	[60-70]	?	2	1	
40926	85504905	Caucasian	Female	[40-50]	?	1	3	
42570	77586282	Caucasian	Male	[80-90]	?	1	6	
62256	49726791	AfricanAmerican	Female	[60-70]	?	3	1	
73578	86328819	AfricanAmerican	Male	[60-70]	?	1	3	
77076	92519352	AfricanAmerican	Male	[50-60]	?	1	1	
84222	108662661	Caucasian	Female	[50-60]	?	1	1	
89682	107389323	AfricanAmerican	Male	[70-80]	?	1	1	
148530	69422211	?	Male	[70-80]	?	3	6	

Extract Table Using Examples

Load

Transform Data

Cancel

encounter_id

patient_nbr

race

gender

age

weight

admission_type_id

discharge_disposition_id

admission_source_id

time in hospital

99986574	62487468	Caucasian	Male	[80-90]	?	1	1		
104772090	80559405	Caucasian	Male	[60-70]	?	1	1		
118450434	54819396	Caucasian	Male	[80-90]	?	1	1		
122355276	40724406	Caucasian	Male	[80-90]	?	1	1		
124231890	67547241	Caucasian	Male	[70-80]	?	1	1		
131068938	57250782	Caucasian	Female	[80-90]	?	1	1		
133044396	109440261	Caucasian	Male	[70-80]	?	1	1		
133872540	64126665	Caucasian	Male	[90-100]	?	1	1		
134045556	98250849	Caucasian	Female	[60-70]	?	1	1		
134604762	81844920	Caucasian	Male	[80-90]	?	1	1		
135453822	66024432	Caucasian	Male	[80-90]	?	1	1		
136452696	40987404	Caucasian	Female	[70-80]	?	1	1		
137897016	55152873	Caucasian	Male	[30-40]	?	1	1		
138309402	109934172	Caucasian	Female	[80-90]	?	1	1		
138365496	39365073	Caucasian	Female	[80-90]	?	1	1		
139287624	50404662	Caucasian	Female	[60-70]	?	1	1		
139593378	105096825	Caucasian	Female	[60-70]	?	1	1		
140311524	50945364	Caucasian	Male	[80-90]	?	1	1		
142017078	61500159	Caucasian	Female	[80-90]	?	1	1		
142541070	97094556	Caucasian	Male	[60-70]	?	1	1		
142554858	40605516	Caucasian	Female	[70-80]	?	1	1		
142854006	57902751	Caucasian	Female	[50-60]	?	1	1		
142883472	75079107	Caucasian	Male	[60-70]	?	1	1		
144625980	58559157	Caucasian	Female	[70-80]	?	1	1		
144976572	33662754	Caucasian	Female	[50-60]	?	1	1		
146015514	111333276	Caucasian	Female	[80-90]	?	1	1		

Name

diabetic_data

Storage mode

Import

Data refreshed

9/13/2026, 8:44:07 PM

Search

diabetic_data

A1Cresult

acarbose

acetoexamide

Σ admission_source_id

Σ admission_type_id

age

change

chlorpropamide

citoglipton

diabetesMed

diag_1

diag_2

diag_3

Σ discharge_disposition_id

Σ encounter_id

examide

gender

glimepiride

glimepiride-pioglitazone

glipizide

nlinizide-metformin

Data Visualization Using Power BI

The **diabetic healthcare dataset** was imported into Power BI Desktop to create interactive visualizations for analyzing **patient readmissions, patient demographics, hospital stay duration, diabetes medication usage, A1C test results, and admission patterns.**

Visualization 1 — Healthcare KPI Cards: Card

Displays **Total Encounters, Readmitted Encounters, Readmission Rate, and Average Hospital Stay**, providing a quick overview of important healthcare metrics.

Visualization 2 — Readmission Distribution: Donut Chart

Shows the **distribution of patients based on their readmission status**, including patients who were not readmitted, readmitted within 30 days, and readmitted after 30 days.

Visualization 3 — Readmission by Age Group: Stacked Column Chart

Compares **patient encounters across different age groups and readmission categories** to identify age-related readmission patterns.

Visualization 4 — Readmission by Gender: Horizontal Bar Chart

Displays the **number of patient encounters across genders and readmission categories** to analyze differences in readmission patterns.

Visualization 5 — A1C Result Distribution: Donut Chart

Shows the **distribution of patient encounters based on A1C test results**, helping analyze diabetes-related laboratory testing patterns.

Visualization 6 — Average Hospital Stay by Age Group: Horizontal Bar Chart

Compares the **average hospital stay across different age groups** to identify variations in hospitalization duration.

Visualization 7 — Hospital Stay by Readmission: Stacked Column Chart

Represents the **distribution of readmission categories across different hospital stay durations**, helping analyze the relationship between length of stay and readmission.

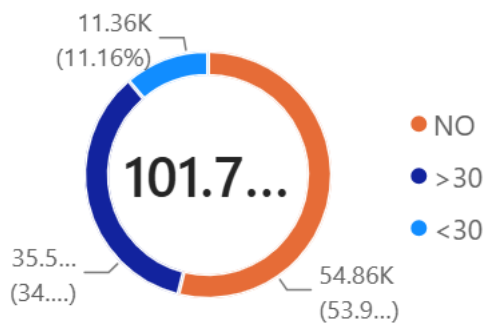
Visualization 8 — Diabetes Medication vs Readmission: Stacked Column Chart

Compares **patient readmission categories based on diabetes medication usage** to identify differences in readmission patterns.

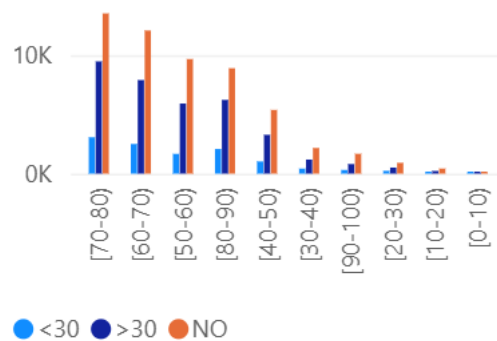
Visualization 9 — Readmission by Admission Type: Bar Chart

Displays the **number of patient encounters across different admission types and readmission categories**, providing insights into admission and readmission patterns.

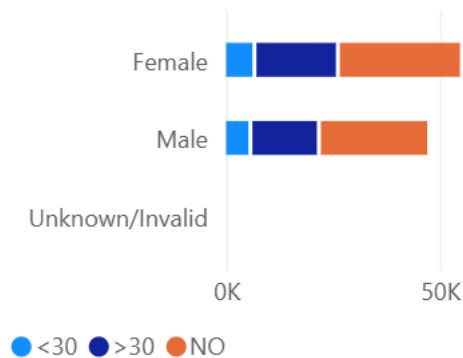
Count of encounter_id
by readmitted



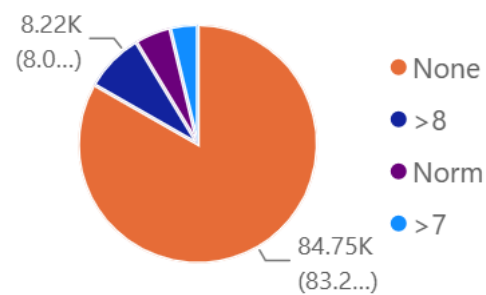
Count of encounter_id
by age and readmitted

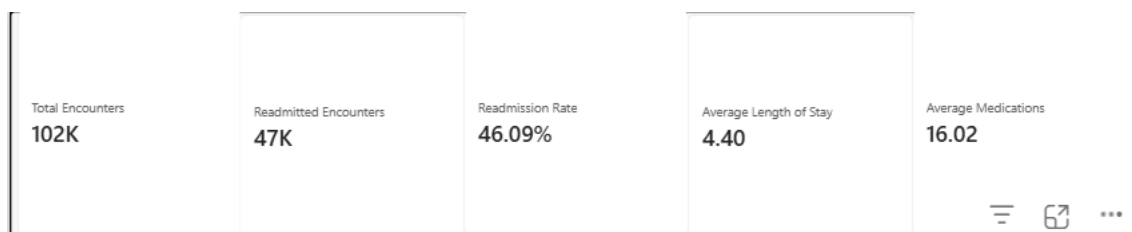
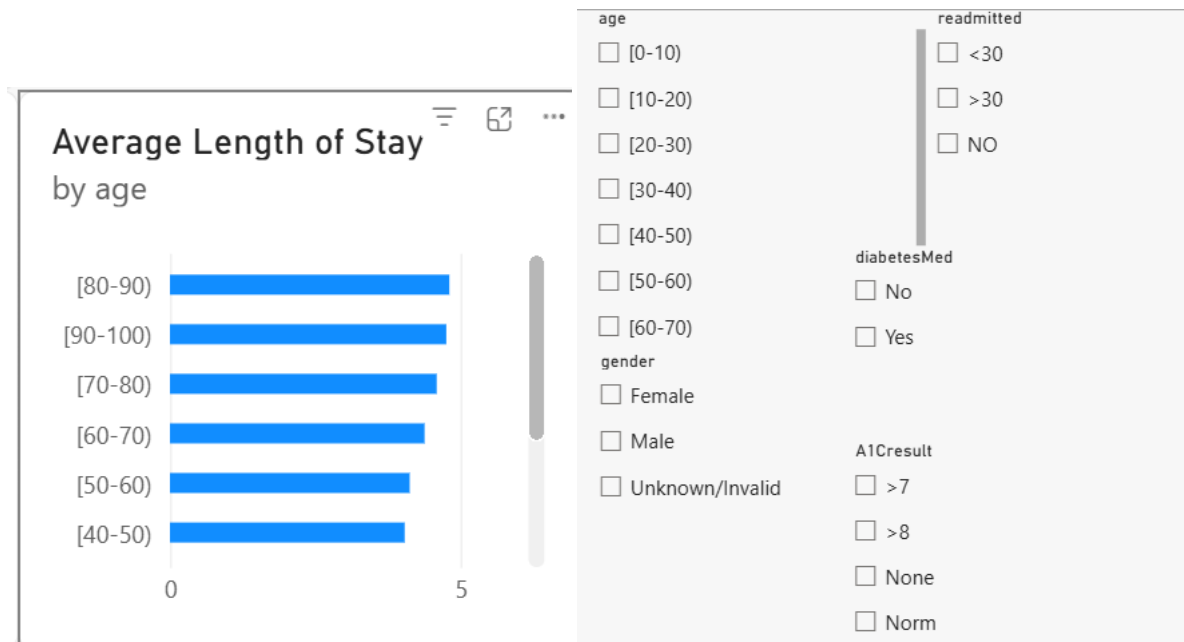


Count of encounter_id
by gender and readmitted

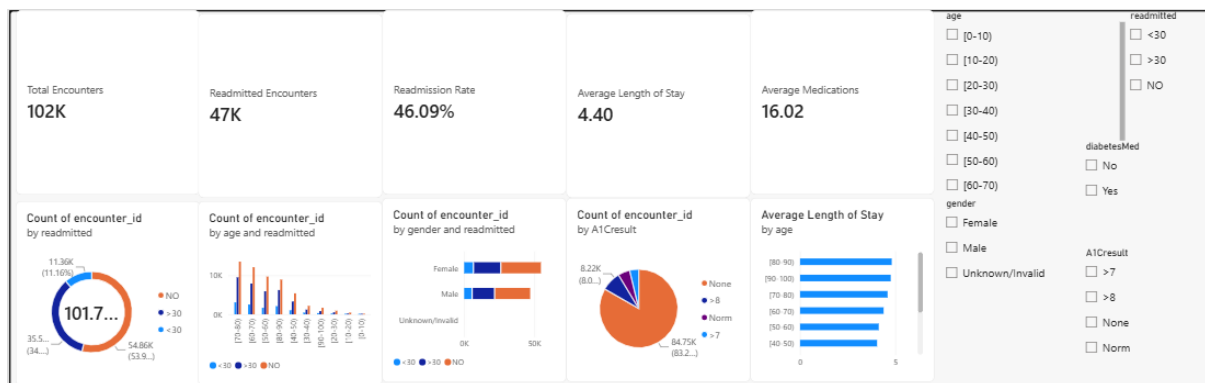


Count of encounter_id
by A1Cresult



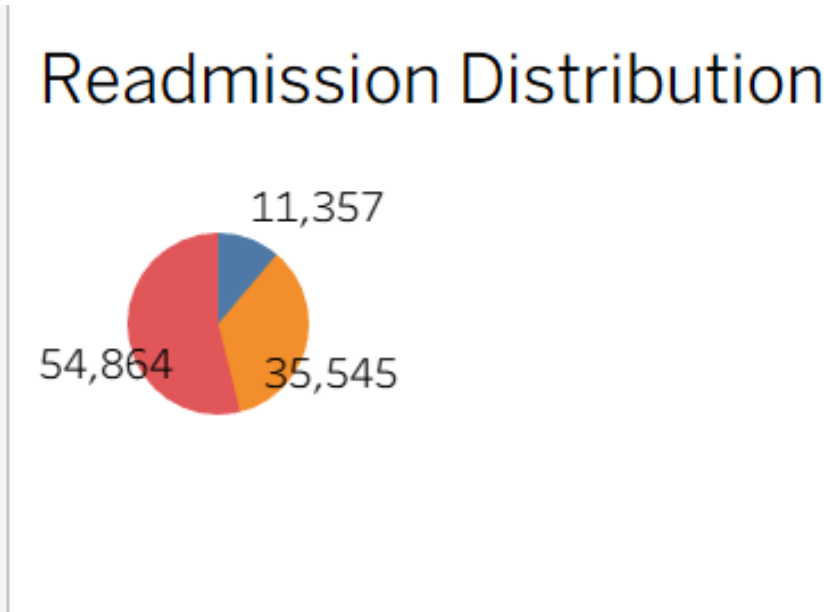
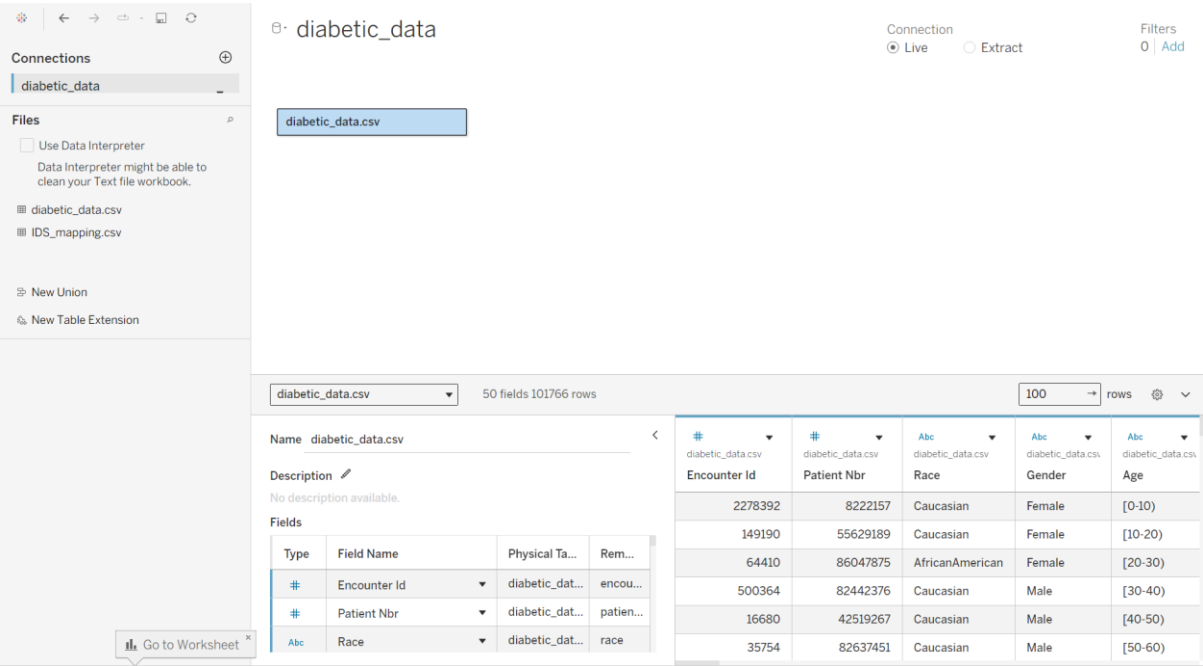


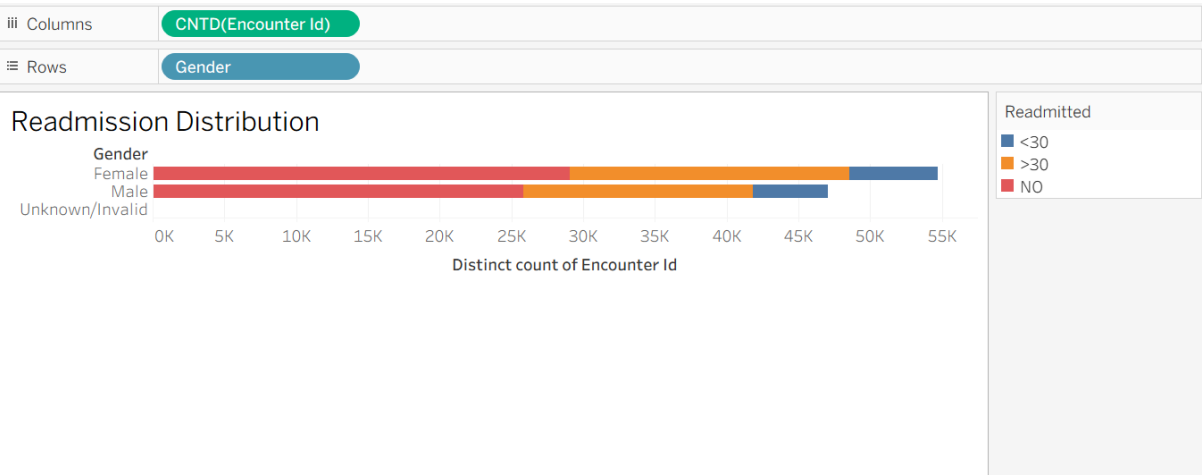
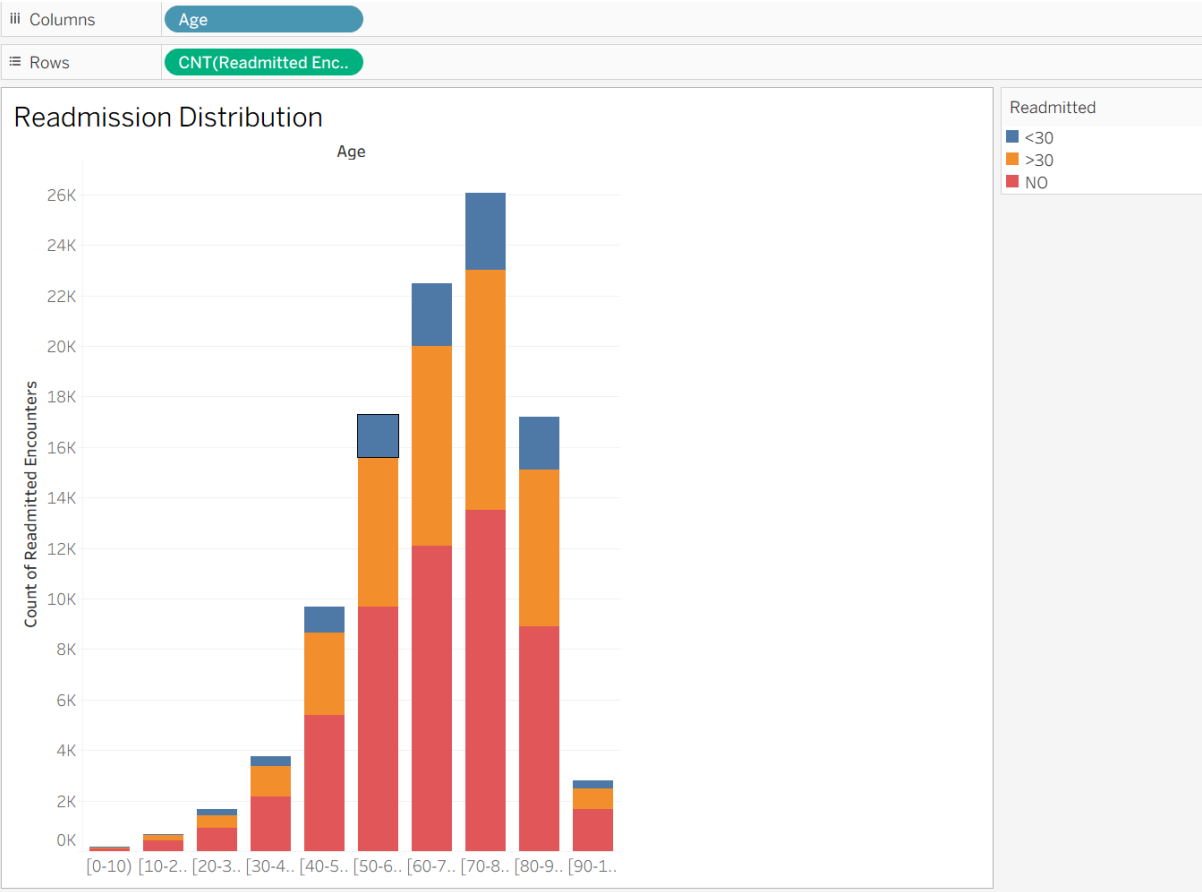
HEALTH CARE ANALYSIS POWER BI DASHBOARD



5. Tableau Dashboard

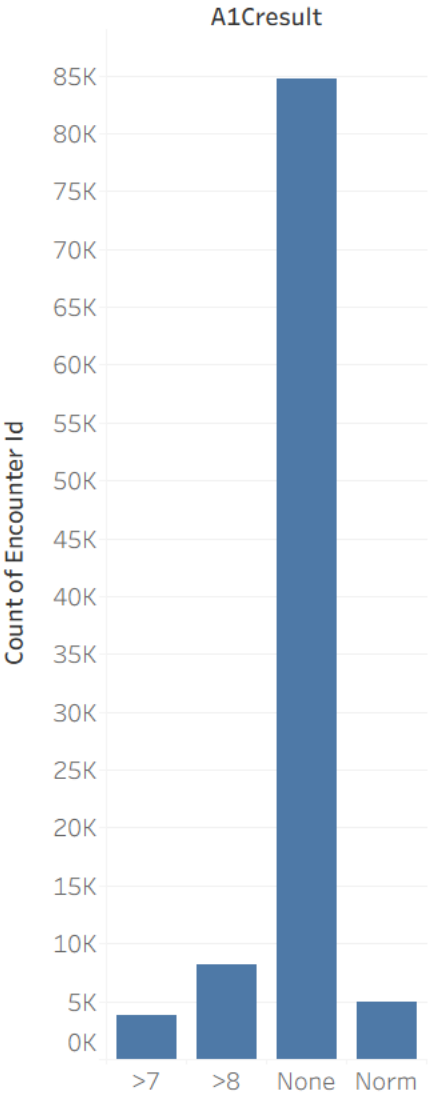
Created visuals in Tableau in separate sheets and created a dashboard



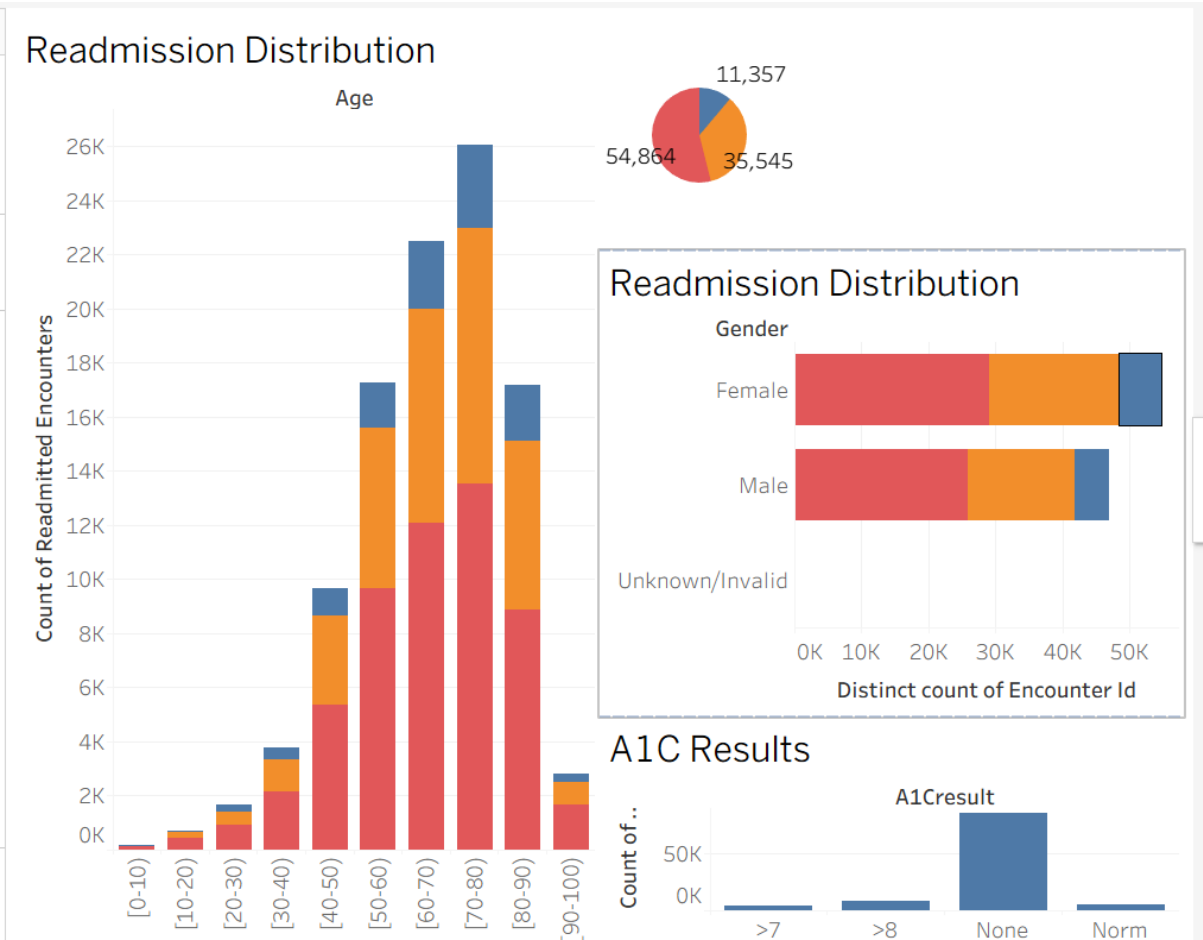


iii Columns	A1Cresult
Rows	CNT(Encounter Id)

A1C Results



HEALTH CARE ANALYSIS TABLEAU DASHBOARD



Result: